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Agency Name:	National Science Foundation
Agency Tracking Number:	2612659
Organization:	Office of Polar Programs (OPP)
NSF Program:	ANS-Arctic Natural Sciences
PI/PD:	Maglio, Benjamin
Application Title:	Collaborative Research: Characterizing Spatiotemporal Heterogeneity of Carbon Cycling Across Ice-wedge Permafrost Landscapes and Projecting Vulnerability to Change

Review 1

Rating:

Very Good

Review:

Summary

Proposal Synopsis

The goal of this proposal is to assess how spatial heterogeneity affects prediction uncertainty across Arctic tundra. The proposal aims to improve predictions of the permafrost carbon-clim feedback in ice-wedge polygon tundra by integrating new field measurements (seasonal and interannual CO2/CH4 chamber fluxes on the Alaskan North Slope) with adaptations of the DV DOS-TEM model to include subgrid parameterization and surface water layer representation.

Intellectual Merit

Strengths

The program encourages proposals that test hypotheses leading to new understanding of the Arctic, including those that synthesize and analyze existing data. The proposal tests three hypothesis and science questions that will test our understanding of how the spatial scale of Terrestrial Biosphere Models (TBMs) are too coarse to represent the CO2 and CH4 fluxes from wedge polygon tundra, which covers about half of the Alaskan Coastal Plain.

They propose to conduct extensive fieldwork to measure CO2 and CH4 fluxes using chambers and other observations linked to the drivers of these fluxes. The subgrid tiling and surface-v layer developments in the model are evaluated against field data structured around the same classification hierarchy (landform/geomorphology/ecotype) used in the model. They will leverage data from the existing NIRPO observatory and other data from Barrow and Atkasuk from NGEE and other ARCSS projects (letters of support provided). They will publicly archive their field data at the Arctic Data Center and they have a safety plan for fieldwork.

Weaknesses

They propose to run the model at coarse (0.5°), moderate ($1\text{--}4\text{ km}^2$), and fine ($<1\text{ km}^2$) resolutions and comparing them against each other and against independent data. However, they do not explain how $\sim 1\text{ km}^2$ resolution is comparable to the ice wedge polygons on the scale of 10 s m^2 . None of the proposed fieldwork and data collection can match any of these scales. For example, ice chambers are $< 1\text{ m}^2$, the ice wedge polygons are 10 s m^2 . Including more information on how that scale gap will be addressed would make the proposal a lot stronger. They do not discuss plans to evaluate the updated model carbon fluxes have a regional impact that matches any larger scale data (e.g. atmospheric observations). They say they will use data from comparison sites but they don't specify how comparable those datasets are in temporal coverage, methodology or scale.

The DVM-DOS-TEM model represents the soil column as a 1-D stack of layers. There does not appear to be any 3d representation of the ice wedge polygons. The surface-water development explicitly "maintains a one-dimensional hydrological model," with only a mention that it "would enable further advancement or coupling to lateral flow two-dimensional hydrological models" as a future possibility. Lateral transport for water and carbon (drainage through degrading IW troughs) is key to resolving carbon fluxes in polygon tundra seems like a real conceptual limitation.

On a writing perspective, I'm confused by what the proposers mean by fine and coarse scale for the model simulations. I expected to see more explicit distances discussed. It would have been easier for me to see how the in situ field observations will be used to improve the model.

Similarly, I didn't follow how the proposers define wet vs moist. How wet is wet? These are classifications with no explicit soil moisture or other metric definitions. It would be good to have an explicit discussion of the variables that will be measured and how they are then defined.

Broader Impacts

Strengths

They have strong broader impacts that I really like.

They have included explicit funding to train undergraduate and postdoctoral researchers in a challenging field environment. The postdoc mentoring plan is good and they have a good safe outlined. They will work with the Data Nuggets team (which started as an NSF project) to create classroom activities for K-12 levels based on the data collected/generated as part of this project. They propose concrete educational activities for both the fieldwork and the model. They will work with teachers to build basic data-analysis skills and give them confidence in how to use them to develop class material. They will update an open source model and share all data publicly on the Arctic Data Center.

The PI is a relatively new PI and he has assembled a good team with lots of experience that can advise him.

Weaknesses

None.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

The proposal includes a dedicated, PI-signed "Safe and Inclusive Working Environment" plan (pp. 98–100), submitted as a supplementary document per NSF PAPPG Chapter II.E.9, covering the Deadhorse, AK / NIRPO field site. There is a lot of detail and good plan should anything occur. Based on my own fieldwork experiences, I am pleased with this plan and think having that second contact (PI's supervisor) available for any project members is particularly important.

Summary Statement

Overall a very good proposal but with some gaps in the link between scales. I think there needs to be a way to assess the model changes on a regional scale. There is also no discussion of lateral transport of carbon or water, which seems like an oversight given the topic.

Review 2

Rating:

Poor

Review:

Summary

Proposal Synopsis

Intellectual Merit

Strengths

Improving the representation of tundra carbon dynamics in land surface models remains a critical research priority. The development of an open-source modeling framework is particularly valuable, as it will facilitate broader community engagement, model development, and reproducibility. A particular challenge is understanding how model spatial resolution influences simulated CO_2 and CH_4 fluxes. Although previous studies have demonstrated that resolution can substantially affect the representation of hydrologic gradients, vegetation heterogeneity, and permafrost processes, considerable uncertainty remains regarding the scales at which key carbon cycling mechanisms must be resolved. Further investigation is needed to quantify how aggregation of scale landscape features influences modeled carbon fluxes and to identify optimal resolutions for accurately projecting Arctic ecosystem responses to climate change.

Weaknesses

The proposal seeks to characterize the environmental controls on carbon cycling across different landforms, geomorphic units, and ecotypes of the Arctic Coastal Plain (ACP), with the ultimate goal of improving regional estimates of CO₂ and CH₄ fluxes. A central objective is to evaluate how carbon fluxes respond to spatial scale and landscape heterogeneity, particularly when segmented into hierarchical classification units. While this is an important question, it is not entirely novel. Previous studies, including the work of Lara et al. (2020), demonstrated that local tundra heterogeneity can substantially influence regional carbon dynamics.

For the data synthesis, a major source of uncertainty is the lack of clarity regarding the quantity, spatial distribution, and quality of carbon flux datasets that will be synthesized. The proposal emphasizes integration of existing observations across the ACP, yet it remains unclear how many studies will contribute data, what geographic regions they represent, and whether sufficient observations exist to characterize environmental variability across such a large and heterogeneous landscape.

An additional challenge is that existing datasets have often been collected using different methodologies, sampling protocols, instrumentation, and temporal frequencies. Standardizing the datasets within a common hierarchical framework of landforms, geomorphic units, and ecotypes will likely be difficult and may introduce substantial uncertainty.

Many vegetation, thaw depth, water table, and geomorphic datasets were collected during different years, seasons, and climatic conditions. Consequently, observed differences among sites reflect short-term environmental variability rather than long-term ecosystem change. This temporal mismatch complicates efforts to identify consistent controls on interannual CO₂ and CH₄ variability. Furthermore, many historical datasets lack key environmental variables required for robust model parameterization including soil moisture, soil temperature, active layer dynamics, water table depth, and seasonal surface water extent. Without these measurements, it may be difficult to establish mechanistic relationships between environmental drivers and carbon fluxes. The proposal does not adequately constrain model parameters. The greatest concern is the uncertainty surrounding how much suitable data are actually available and whether limited site-level observations can realistically support forecasts of carbon dynamics across the entire North Slope.

The proposal highlights the importance of seasonality in controlling Arctic carbon fluxes, yet the temporal sampling strategy remains insufficiently described.

The project appears to rely heavily on a previously collected dataset from the study site that was acquired primarily during peak growing season (Kade et al 2024). Although the proposal indicates that new seasonal measurements will be collected, it does not specify sampling frequency (weekly, biweekly, monthly). These details are particularly important because carbon fluxes in Arctic systems can vary dramatically over short timescales and may contribute substantially to annual carbon budgets. Without adequate temporal coverage, important seasonal controls on carbon exchange may be overlooked.

In regards of the site and scaling: The proposal aims to represent carbon dynamics across multiple hierarchical levels, including landforms, geomorphic units, and polygon-scale features. However, the proposed field site appears to provide only limited representation of the environmental diversity found across the ACP. Several concerns arise including (i) the fact that this site encompasses a relatively narrow range of landforms and geomorphic settings, (ii) the location is situated at the center of Prudhoe Bay, an area affected by infrastructure development, networks, dust deposition, and localized permafrost degradation associated with anthropogenic disturbance and urban heat island effects. (iii) The proposed three transects are clustered within a relatively small area and therefore may not adequately capture the spatial heterogeneity characteristic of the broader ACP. As a result, relationships observed at chamber locations or survey plots may not scale predictably to larger ecotype or landform classes, making regional extrapolation challenging.

The proposed modeling framework relies heavily on field observations for calibration and validation. However, the available observations appear limited relative to the spatial and temporal scales at which model predictions are intended.

Broader Impacts
Strengths

This project seeks to advance understanding of not only regional carbon cycling but helps improve representation of the arctic system in land models. The funds will train new students and postdocs while supporting early career and senior scientists. The proposed broader impacts on incorporating lessons plans based on geospatial data and study products into K-12 is great.

Weaknesses

the broader impacts are fair and align with NSF.

Additional Solicitation-specific Review Criteria
Strengths & Weaknesses
Summary Statement

Improving the representation of tundra carbon dynamics in open-source land surface models is a critical research priority that will enhance transparency, reproducibility, and community-driven model development. Although previous studies have demonstrated that model resolution strongly influences simulated CO₂ and CH₄ fluxes, important uncertainties remain regarding the scale needed to accurately represent Arctic landscape heterogeneity and carbon dynamics. The project seeks to integrate, test, and upscale fine-scale carbon flux observations to regional estimates through a hierarchical framework of landscape units, including landforms, geomorphic classes, and ecotypes. However, the general approach is not novel, as previous studies have similar hierarchical landscape classifications to link local-scale heterogeneity with regional carbon dynamics across Arctic ecosystems. In addition, the approach contains some challenges, particularly the disconnect between the ambitious regional-scale modeling objectives and the limited carbon cycling observations. Significant uncertainty remains regarding the quantity, consistency, scale, representativeness, and seasonal coverage of the datasets that will be the base for running the model.

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